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Climate Change and Sustainable Development in Pakistan: An Analysis of the 2030 Agenda

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INTRODUCTION

The United Nations, along with its member countries, has embraced the 2030 agenda, a comprehensive framework comprising 17 Sustainable Development Goals (SDGs). This agenda outlines a set of 169 specific targets aimed at guiding global development efforts. Located at the crossroads of South Asia, Pakistan is a country with rich cultural diversity, an increasing population, and a dynamic economy. Yet, like many developing nations, it grapples with a diverse challenges ranging from poverty, inequality, and education deficits to environmental degradation and healthcare disparities. The primary objectives of this paper revolve around examining the impact of climate change on the Environmental Group (SDG 6,13,14 and 15) of SDGs. This investigation will employ rigorous statistical analyses, specifically focusing on correlation, covariance, and p-values, to provide a quantitative understanding of the relationships involved.

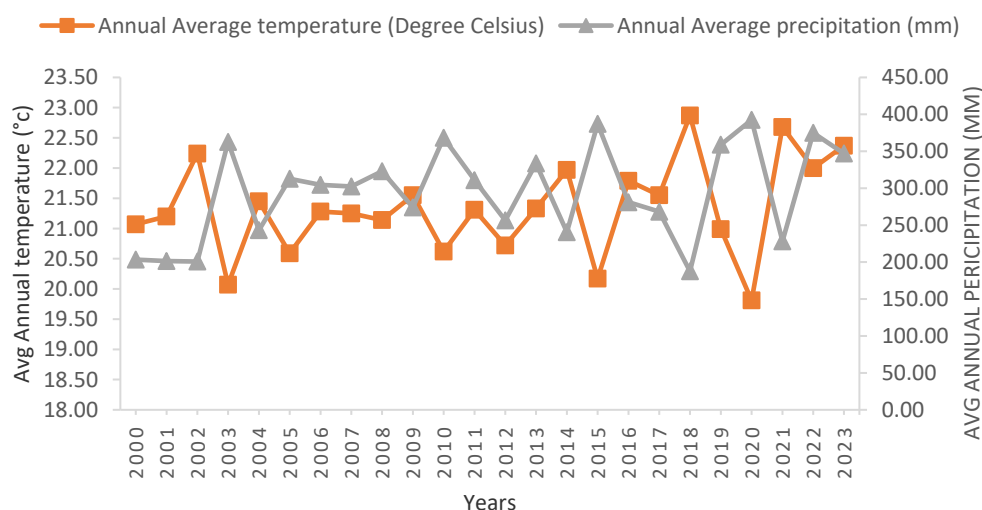


Figure 13 Annual average chnagne in Temperature and Precipitation

MATERIALS AND METHODS

The analytical methods used here are novel, in their use within the tenets of biomimicry and nature-inspired systems thinking. The use of correlation and covariance to examine the

interplay between climate variables and SDG indicators mimics ecological systems thinking — recognizing that environmental change manifests not through isolated variables, but through interdependent dynamics. This systems-based perspective advocates for an understanding of natural environments as complex, adaptive systems. The observed feedback loops — such as the inverse relationship between precipitation and water stress, or the resilience of mangrove ecosystems — mirror the adaptive and self-regulating traits found in natural ecological systems (Cetin et al., 2024). Figure 2 shows the analytical framework in a flow diagram.

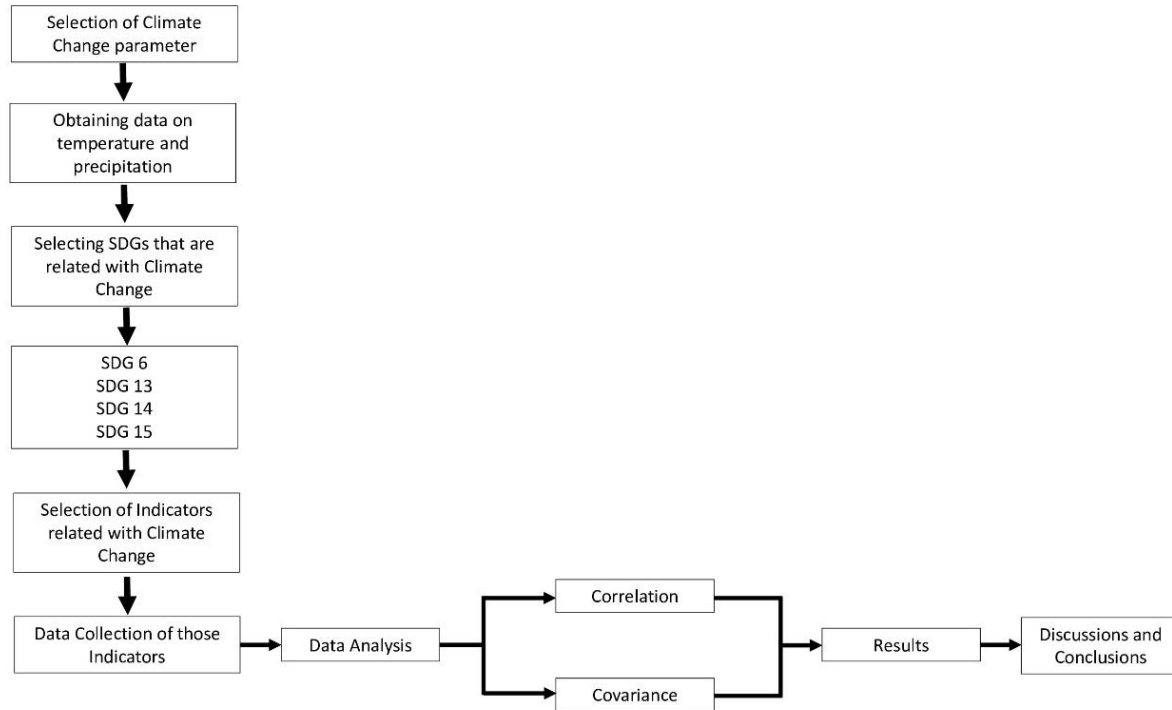


Figure 14 Conceptual Framework of Methodology

The methods here assess the impact of climate change on specific SDGs by analyzing climate data, particularly temperature and precipitation. It starts with identifying relevant SDGs (SDG 6, 13, 14, and 15) and selecting indicators linked to climate effects. Data for these indicators is collected and analyzed through correlation and covariance to determine relationships and trends.

RESULTS AND DISCUSSION

The analysis shows that temperature and precipitation changes have distinct effects on water and ecosystem indicators. Seasonal lake areas are highly responsive, expanding significantly with increased precipitation, while permanent lakes are relatively stable. Mangrove areas show moderate adaptability to both temperature and precipitation changes. Water stress is strongly affected, increasing notably with higher temperatures and reduced precipitation, highlighting the risk of water scarcity. Meanwhile, water efficiency improves moderately, likely due to adaptive practices in response to climate pressures. These insights emphasize the need for targeted climate adaptation strategies for water resources and ecosystems.

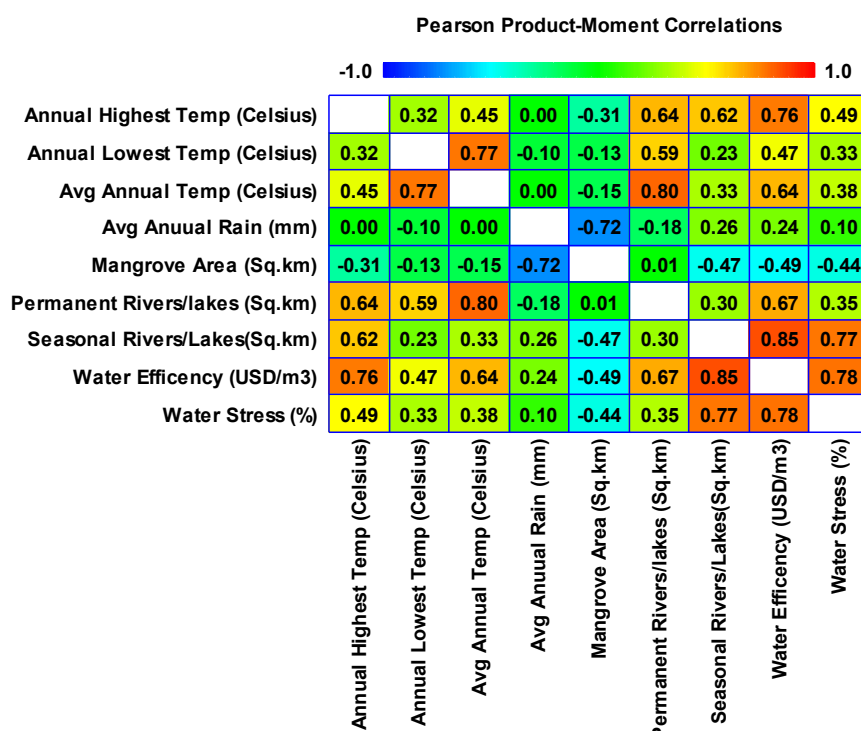


Figure 15 Pearson Product-Moment correlation chart of SDG6 Indicators and Climate change Parameters

Table 8 Correlation and P-value of SDG6 indicators and Climate Change Parameters

	Correlations								
	Annual Highest Temp (°C)	Annual Lowest Temp (°C)	Avg Annual Temp (°C)	Avg Annual Rain (mm)	Mangrove Area (Sq. km)	Permanent Rivers/Lakes (Sq. km)	Seasonal Rivers/Lakes (Sq. km)	Water Efficiency (USD/m ³)	Water Stress (%)
Annual Highest Temp (°C)	Cor.	0.3165	0.4528	0.002	-0.3053	0.6423	0.6164	0.7593	0.4928
	Size P	20 0.1739	20 0.045	20 0.9933	20 0.1905	20 0.0023	20 0.0038	20 0.0001	20 0.0273
Annual lowest Temp (°C)	0.3165	Cor.	0.772	-	-0.1349	0.5923	0.2263	0.474	0.3347
	20 0.1739	Size P	20 0.0001	0.1017 20 0.6696	20 0.5706	20 0.0059	20 0.3374	20 0.0347	20 0.1492
Avg Annual Temp (°C)	0.4528	0.772	Cor.	-	-0.1507	0.7979	0.3332	0.6417	0.3767
	20 0.045	20 0.0001	Size P	0.0047 20 0.9845	20 0.526	20 0	20 0.1511	20 0.0023	20 0.1016
Avg Annual Rain (mm)	0.002	-	-	Cor.	-0.7206	-0.1816	0.2577	0.2356	0.0954
	20 0.9933	0.1017 20 0.6696	0.0047 20 0.9845	Size P	20 0.0003	20 0.4436	20 0.2727	20 0.3173	20 0.689
Mangrove Area (Sq. km)	-	-	-	-	Cor.	0.0056	-0.4678	-0.492	-
	0.3053	0.1349	0.1507	0.7206	Size	20	20	20	0.4424
	20	20	20	20		20	20	20	20

	0.1905	0.5706	0.526	0.0003	P	0.9812	0.0375	0.0276	0.0508
Permanent Rivers/Lakes (Sq. km)	0.6423	0.5923	0.7979	-	0.0056	Cor.	0.3008	0.669	0.3473
	20	20	20	0.1816	20	Size	20	20	20
	0.0023	0.0059	0	0.4436	0.9812	P	0.1975	0.0013	0.1335
Seasonal Rivers/Lakes (Sq. km)	0.6164	0.2263	0.3332	0.2577	-0.4678	0.3008	Cor.	0.8453	0.7725
	20	20	20	20	20	20	Size	20	20
	0.0038	0.3374	0.1511	0.2727	0.0375	0.1975	P	0	0.0001
Water Efficiency (USD/m³)	0.7593	0.474	0.6417	0.2356	-0.492	0.669	0.8453	Cor.	0.7801
	20	20	20	20	20	20	20	Size	20
	0.0001	0.0347	0.0023	0.3173	0.0276	0.0013	0	P	0
Water Stress (%)	0.4928	0.3347	0.3767	0.0954	-0.4424	0.3473	0.7725	0.7801	Cor.
	20	20	20	20	20	20	20	20	Size
	0.0273	0.1492	0.1016	0.689	0.0508	0.1335	0.0001	0	P

Cor.: Correlation

P.: P-value

CONCLUSION

Climate change, particularly temperature and precipitation shifts, has varying impacts on water and ecosystem indicators. Seasonal lake areas and water stress are highly climate-sensitive, while permanent lakes and mangroves show resilience. Water efficiency improvements may reflect adaptive responses rather than lasting solutions. This study suggests that climate-adaptive water management, especially for stress-prone areas, is crucial. Practical recommendations include enhancing seasonal water storage and pursuing sustainable efficiency measures

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